Tutorial-3

A mobile service provide wants to classify SMS messages as spam or not spam using a single layer neural network with 5 hidden neurons in the layer.

x_1 = specific works and phrases

x_2 = links and Attachments

x_3 = bias.

x_1	x_2	x_3	Target (y)
4	2	1	1
1	0	1	0
6	3	1	1
2	1	1	0

	w_1	w_2	b
h_1	0.1	-0.2	1
h_2	0.3	-0.1	1
h_3	0.2	0.4	1
h_4	0.1	0.3	1
h_5	-0.2	0.4	1

$v_1 = 0.1$

$v_2 = 0.2$

$v_3 = -0.3$

$v_4 = -0.1$

$v_5 = 0.4$

$$\boxed{z_1 = x_1 w_1 + x_2 w_2 + b}$$

$z_1 = 4(0.1) + 2(0.2) + 1 = 1$

$z_2 = 4(0.3) + 2(-0.1) + 1 = 2$

$z_3 = 4(0.2) + 2(0.4) + 1 = 2.6$

$z_4 = 4(0.1) + 2(0.3) + 1 = 2$

$z_5 = 4(-0.2) + 2(0.4) + 1 = 1$

Hidden neurons

$h_1 = \sigma(z_1) = \frac{1}{1+e^{-z}} = 0.731$

$h_2 = \sigma(z_2) = 0.880$

$h_3 = \sigma(z_3) = 0.930$

$h_4 = \sigma(z_4) = 0.880$

$h_5 = \sigma(z_5) = 0.731$

$$\boxed{z_{out} = h_1 v_1 + h_2 v_2 + h_3 v_3 + h_4 v_4 + h_5 v_5}$$

$= 0.73 \times 0.1 + 0.88 \times 0.2 + 0.93 \times (-0.3) + 0.88 \times (-0.1) + 0.73 \times (0.4)$

$$\boxed{z_{out} = 0.174}$$

$\hat{y} = \sigma(z_{out}) = \frac{1}{1+e^{-0.174}} = 0.543$

$\hat{y} \geq 0.5 \Rightarrow 1$ $\boxed{\hat{y} = 1}$ $\boxed{y = 1}$

$\therefore \boxed{y = \hat{y}}$

Sample 2: (1, 0, 1)

$z_1 = 1(0.1) + 0(-0.2) + 1 = 1.1$

$z_2 = 1(0.3) + 0(-0.1) + 1 = 1.3$

$z_3 = 1(0.2) + 0(0.4) + 1 = 1.2$

$z_4 = 1(0.1) + 0(0.3) + 1 = 1.1$

$z_5 = 1(-0.2) + 0(0.4) + 1 = 0.8$

Hidden neurons:

$h_1 = 0.750$

$h_2 = 0.785$

$h_3 = 0.768$

$h_4 = 0.750$

$h_5 = 0.689$

$z_{out} = 0.750 \times 0.2 + 0.785 \times 0.2 + 0.768 \times (-0.3) + 0.750 \times (-0.1) + 0.689$

$\boxed{z_{out} = 0.2022}$

$\hat{y} = 0.550$

$\hat{y} = 1 \quad y = 0$

$\boxed{y \neq \hat{y}}$

Updating weights

$y = 0 \quad \hat{y} = 0.55 \quad \eta = 0.1$

$\eta(y - \hat{y}) = 0.1(0 - 0.55) = -0.055$

$\boxed{v_i(new) = v_i(old) + \eta(y - \hat{y}) h_i}$

$v_1 new = 0.1 + (-0.055)(0.750) = 0.05875$

$v_2 new = 0.2 + (-0.055)(0.785) = 0.1568$

$v_3 new \Rightarrow 0.3 + (-0.055)(0.768) = -0.34224$

$v_4 new \Rightarrow -0.1 + (-0.055)(0.750) = -0.14125$

$v_5 new \Rightarrow 0.4 + (-0.055)(0.689) = 0.362105$

Sample 3 (6, 3, 1)

$z_1 = 6(0.1) + 3(-0.2) + 1 = 1$

$z_2 = 6(0.3) + 3(-0.1) + 1 = 2.5$

$z_3 = 6(0.2) + 3(0.4) + 1 = 3.4$

$z_4 = 6(0.1) + 3(0.3) + 1 = 2.5$

$z_5 = 6(-0.2) + 3(0.4) + 1 = 1$

Hidden neurons:

$h_1 = 0.731$

$h_2 = 0.924$

$h_3 = 0.967$

$h_4 = 0.924$

$h_5 = 0.731$

$z_{out} = 0.731(0.1) + 0.924(0.2) + 0.967(-0.3) + 0.924(-0.1) + 0.731 \times 0.3$

$\boxed{z_{out} = 0.1678}$

$y = 0.541 \Rightarrow y = 1 \quad \hat{y} = 1$

$\boxed{y = \hat{y}}$

Sample 4: (2, 1, 1)

Hidden neurons:

$z_1 = 2(0.1) + 1(-0.2) + 1 = 1$ $\qquad h_1 = 0.731$

$z_2 = 2(0.3) + 1(-0.1) + 1 = 1.5$ $\qquad h_2 = 0.817$

$z_3 = 2(0.2) + 1(0.4) + 1 = 1.8$ $\qquad h_3 = 0.858$

$z_4 = 2(0.4) + 1(0.3) + 1 = 1.5$ $\qquad h_4 = 0.817$

$z_5 = 2(-0.2) + 1(0.4) + 1 = 1$ $\qquad h_5 = 0.731$

$z_{out} = 0.731(0.1) + 0.817(0.2) + 0.858(-0.3) + 0.817(0.1) + 0.731(0$

$\boxed{z_{out} = 0.1898}$

$z_{out} = 0.547 \Rightarrow \hat{y} = 1 \quad y = 0$

$\therefore \boxed{y \neq \hat{y}}$

Updating weights:

$y = 0 \quad \hat{y} = 0.547, \quad \eta = 0.1$

$\eta(y - \hat{y}) = 0.1(0 - 0.547) \Rightarrow -0.0547$

$v_1 = 0.1 + (-0.0547)(0.731) = 0.060$

$v_2 = 0.2 + (-0.0547)(0.817) = 0.155$

$v_3 = 0.3 + (-0.0547)(0.858) = -0.346$

$v_4 = -0.1 + (-0.0547)(0.817) = -0.144$

$v_5 = 0.4 + (-0.0547)(0.731) = 0.360$

Final weights